Jeremy Barry

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EDUCATION

William & Mary | *Graduating December 2027 | Williamsburg, VA*
B.S: Data Science and Mathematics, Finance Minor

- ***Major GPA****: 4.00* | ***GPA***: *3.83*
- Relevant Coursework- Applied Machine Learning, Automation & Workflows, Multivariable Calculus, Trustworthy AI, Databases
- Clubs and Activities: Tribe Innovation, AI Club, Club Pickleball, Google Student Developer Club, Sigma Pi Scholarship Chair

WORK EXPERIENCE

Data Science Intern | *May 2026- August 2026 | Washington D.C*
NT Concepts

- Built, trained, and benchmarked detection models using high-fidelity synthetic datasets engineered to replicate diverse adverse weather conditions, improving robustness for edge deployment.
- Developed a generative video pipeline using FLUX.2, ControlNet, and frame interpolation to simulate realistic adversarial weather attacks against fine-tuned YOLO models.
- Created a differentiable rendering framework using PyTorch3D and Mitsuba to identify optimal adversarial patch locations across 3D models and HDR environments, achieving effective attacks with just 6% surface coverage.

AI Engineering Intern | *May 2025- August 2025 | San Francisco Bay Area*
Oaklet (AI-enabled electronic health record for behavioral therapists)

- Architected and developed scalable, HIPAA-compliant AI infrastructure on AWS, building production data pipelines and backend systems for behavioral healthcare LLM applications.
- Researched and prototyped AI solutions with AWS Bedrock that reduced therapist administrative work by 6+ hours per week and supported an estimated $5,800 in additional monthly revenue per clinician.
- Built agentic AI workflows using LangChain, MCP servers, RAG, and tool-calling architectures, enabling models to retrieve clinical context, interact with application systems, and automate multi-step therapist workflows.

Data-Driven Strategy & Quality Improvement Intern | *August 2025 - December 2025| Williamsburg, VA*
Eastern State Hospital

- Leveraged AI to build internal dashboards that revealed previously untracked workforce inefficiencies (e.g. nurse turnover, excessive time in restraint), enabling leadership to implement changes that target root-cause issues.
- Analyzed workforce retention data to identify root causes of turnover across departments and shifts, informing policy changes that reduced turnover by 33%.

ENTREPENERUSHIP & RESEARCH

Zumah | *July 2026 - Present | Founder*

- Built an AI cost-optimization API that dynamically routes prompts across LLMs, compresses context, and leverages KV caching to reduce redundant token usage and costs while maintaining response quality and integrating into existing LLM workflows.
- Launched a successful affiliate program and supported enterprise pilots and benchmarking that demonstrated up to 85% cost savings, helping drive early customer interest and adoption.

Machine Learning Researcher | *August 2026 - Present | William & Mary*

- Developing a machine learning classifier for diabetic retinopathy detection from retinal fundus images, training and evaluating computer vision models on large-scale ophthalmic imaging datasets.
- Investigating topological and theoretical mathematical methods to improve sparse convolutional neural networks, aiming to capture retinal structure efficiently while reducing reliance on transformer-based architectures.

Crawllr | *August 2025- April 2026 | Co-Founder*

- Engineered and launched a full-stack React Native mobile app with real-time geolocation, event routing, authentication, and Google Cloud infrastructure across iOS and Android.
- Won William & Mary's Rocket Pitch competition and led product, growth, and go-to-market execution, launching ambassador and venue-partnership programs that all helped scale the app to 800+ users in the first month.

CUSTOMER SERVICE EXPERIENCE

Head Shucker | *May 2024- July 2025 | Newburyport, MA*
Brine Oyster

- Shucked over 500 oysters per shift in a high-end, fast-paced restaurant, ensuring consistent quality and top-tier guest experience. Managed and trained members of the oyster-shucking team, ensuring efficiency and adherence to restaurant standards.

AWARDS, SKILLS & INTERESTS

Awards/Recognitions: Gold Key Recipient, Dean's List (4x), Class President (4x)
Technical Skills: Python, SQL, PyTorch, MLFlow, Tableau, AWS, Git, Data Modeling, GenAI, Kubernetes, Deep Learning, Excel
Interests: Eating Pizza, Attempting Marathons, Going to Cool Places, Robots